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Age and paleomagnetism of the 1210 Ma Gnowangerup–Fraser dyke swarm, Western Australia, and implications for late Mesoproterozoic paleogeography

Sergei A. Pisarevsky[a,b,c,*], Michael T.D. Wingate[a,b,c,d,e], Zheng-Xiang Li[a,b], Xuan-Ce Wang[a,b], Eric Tohver[c], Christopher L. Kirkland[a,d,e]

[a] Australian Research Council Centre of Excellence for Core to Crust Fluid Systems (CCFS), Australia
[b] The Institute for Geoscience Research (TIGeR), Department of Applied Geology, Curtin University, GPO Box U1987, Perth, WA 6845, Australia
[c] School of Earth and Environment, University of Western Australia, Crawley, WA 6009, Australia
[d] Department of Imaging and Applied Physics, Curtin University, GPO Box U1987, Perth, WA 6845, Australia
[e] Geological Survey of Western Australia, 100 Plain St., East Perth, WA 6004, Australia

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ABSTRACT

Dolerite dykes of the Gnowangerup–Fraser Dyke Suite are subparallel to the southern and southeastern margins of the Yilgarn Craton. We collected oriented samples for paleomagnetic study from 19 dykes along the Phillips and Fitzgerald Rivers and north of Ravensthorpe. Alternating-field (AF) demagnetization revealed a stable two-polarity remanence in 14 dykes, and the primary nature of the magnetic directions is supported by a positive baked-contact test and by rock-magnetic evidence. U–Pb zircon and baddeleyite ages for two dykes confirm that the Gnowangerup–Fraser dykes are part of the 1210 Ma Marnda Moorn Large Igneous Province. The mean paleomagnetic pole, at 55.8° N, 323.9° E, A_{95} = 6.5°, is almost identical to the previously reported VGP of the 1212 Ma Fraser dyke, also supported by a positive baked-contact test. The combined robust paleopole places the West Australian Craton in a near-polar position at 1210 Ma. Comparison with coeval Laurentian paleopoles indicates that Laurentia and Australia were widely separated at that time. We present a paleomagnetically permissible drift model for these two continents between 1210 and 1070 Ma. One dyke yields a stable remanence with a VGP similar to the paleopole for the 755 Ma Mundine Well dykes indicating that this dyke may have been emplaced during the same event at c. 755 Ma. Differences in lengths and shapes of late Mesoproterozoic Apparent Polar Wander Paths of several continents suggests that a large supercontinent did not exist between about 1300 and 1050 Ma. This may have been a transitional time between the final breakup of Nuna and the assembly of Rodinia.

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1. Introduction

The popular concept of supercontinent cycles (e.g., Nance and Murphy, 2013, and references therein) suggests the existence of at least two Precambrian supercontinents, referred to as Nuna (or Columbia) and Rodinia. The times of their assembly and breakup are debated, as are their constituents and configurations (e.g., Hoffman, 1991; Dalziel, 1997; Pisarevsky et al., 2003a; Pesonen et al., 2003; Li et al., 2008; Zhao et al., 2004; Evans and Mitchell, 2011; Zhang et al., 2012; Pisarevsky et al., 2014). The late Mesoproterozoic (1300–1000 Ma) is generally regarded as the time of the breakup of Nuna and/or the subsequent assembly of Rodinia (e.g., Pisarevsky et al., 2014, and references therein). However, late Mesoproterozoic paleomagnetic results are sparse, hence any new reliable and well dated paleomagnetic pole provides important information about this enigmatic period of Earth's history.

Although it is not yet possible to construct a reliable Apparent Polar Wander Path (APWP) for each Mesoproterozoic continent, there are problems that can be addressed with available data. In particular, new paleomagnetic data can be used for testing the very existence of a supercontinent: if the angular distance between two well-dated poles from one continent is significantly different from that between two poles of similar age from another continent, it

* Corresponding author at: School of Earth and Environment, University of Western Australia, Crawley, WA 6009, Australia. Tel.: +61 8 6488 5076.
E-mail addresses: Sergei.Pisarevsky@uwa.edu.au, Sergei.Pisarevskiy@curtin.edu.au (S.A. Pisarevsky).

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is impossible that these two continents moved coherently as parts of a single supercontinent (Evans and Pisarevsky, 2008). However, the opposite is not necessarily true, because even if this angular distance is the same for two continents, these continents could drift differently–two points can be connected by various curves. Hence more paleomagnetic data are required for intervening times. All paleomagnetic reconstructions should also be tested with, and constrained by, geological information.

Previous paleomagnetic studies of the Gnowangerup–Fraser dykes included preliminary results for four dykes (the data were not published in detail, Giddings, 1976), and a well-dated virtual geomagnetic pole (VGP), which was shown to be primary by a positive baked-contact test but which probably did not adequately average paleosecular variation (Wingate et al., 2000, 2002a; Pisarevsky et al., 2003b). In this contribution, we present new paleomagnetic and geochronological data for the c. 1210 Ma Gnowangerup–Fraser dolerite dyke swarm in the southern part of Western Australia (Fig. 1). The results confirm the age of the dykes and their affinity with the Marnda Moorn LIP, provide a high-quality, well-dated paleomagnetic pole for Australia at c. 1210 Ma, and allow us to comment on the reconstructions of continents during the late Mesoproterozoic.

2. Geology and sampling

Ion microprobe (SHRIMP) U–Pb geochronology of mafic dykes from the interior and margins of the Yilgarn Craton led to recognition of the c. 1210 Ma Marnda Moorn large igneous province (Wingate et al., 2005; Wingate and Pidgeon, 2005; Wingate, 2007). The Marnda Moorn LIP consists only of gabbro, dolerite, and diorite dykes (Fig. 1); no large layered intrusions or volcanic rocks are known, although the poorly understood basaltic Morawa Lavas

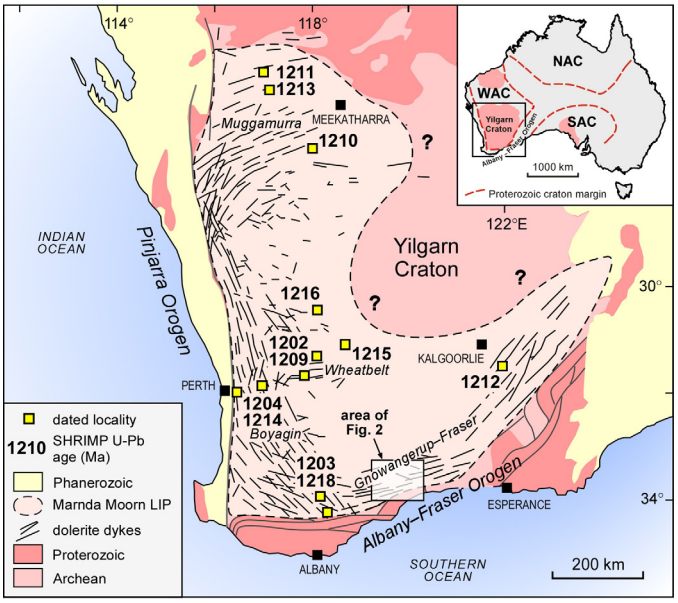


Fig. 1. Simplified map of the Marnda Moorn large igneous province (after Wingate, 2007), showing locations of previously dated samples (U–Pb ages are from Evans, 1999; Qiu et al., 1999; Wingate et al., 2000; Pidgeon and Nemchin, 2001; Pidgeon and Cook, 2003; Rasmussen and Fletcher, 2004; Wingate and Kirkland, 2011; MTD Wingate, unpublished data). Mafic dykes are shown schematically, and very few are dated; the individual dyke swarms (named in italics) may include dykes of different ages. The inset shows the Yilgarn Craton and the relative positions of the West, North, and South Australian Cratons (WAC, NAC, SAC), which are separated by Proterozoic orogenic belts.

Table 1
Paleomagnetic directions and poles for dolerite dykes of the Gnowangerup–Fraser Dyke Suite.

#	Dyke	N/n	Slat (°S)	Slong (°E)	Decl. (°)	Incl. (°)	k	α_{95} (°)	Plat (°N)	Plong (°E)	Dp (°)	D_m (°)
Group 1 (c. 1210 Ma)												
1	RD01	24/21	33°27.747′	120°04.542′	136.7	73.3	37.2	5.3	51.7	334.8	8.5	9.5
2	RD02	11/11	33°26.318′	120°02.019′	143.1	77.5	110.2	4.4	50.7	322.6	7.7	8.2
3	RD03	8/7	33°26.346′	120°01.935′	130.6	65.4	71.4	7.2	50.6	354.0	9.5	11.7
4	RD05	16/13	33°36.364′	119°53.039′	196.1	76.4	24.7	8.5	57.2	286.8	14.6	15.7
5	RD06	18/11	33°36.435′	119°53.165′	138.3	74.0	20.1	10.4	52.1	332.5	16.9	18.8
6	RD07	6/5	33°36.435′	119°53.165′	134.4	79.9	25.7	15.4	45.8	320.0	28.2	29.5
7	RD08[a]	11/9	33°36.485′	119°53.237′	141.8	82.5	12.4	15.2	44.6	299.9	28.9	29.7
8	RD09	15/8	33°36.509′	119°53.290′	143.8	71.4	12.2	12.9	56.6	336.6	19.8	22.6
9	RD11	9/8	33°35.098′	119°52.360′	133.8	78.3	42.2	8.6	47.0	323.9	15.3	16.2
10	RD12[b]	4/4	33°36.753′	119°53.784′	154.9	76.0	158.5	7.3	56.3	319.8	12.4	13.5
11	RD13[c]	16/16	33°36.909′	119°53.935′	53.7	64.6	155.5	3.0	3.6	333.7	3.9	4.8
12	RD15	16/7	33°49.606′	119°15.673′	11.2	-71.9	15.3	15.9	65.8	284.3	24.7	28.0
13	RD17	17/13	33°49.477′	119°15.570′	144.7	72.1	25.0	8.5	56.7	334.0	13.2	15.0
14	RD18	14/14	33°49.477′	119°15.570′	157.0	71.9	32.9	7.0	62.2	326.5	10.9	12.3
15	RD19	12/12	33°49.439′	119°15.558′	148.0	64.8	61.5	5.6	62.6	351.5	7.3	9.0
Mean of 14 dykes					149.0	74.8	122.3	3.6	55.8	323.9	A_{95} = 6.5	
Group 2 (c. 750 Ma?)												
16	RD12[d]	9/8	33°36.753′	119°53.784′	188.6	−40.5	20.7	12.5	32.7	129.3	9.1	15.1
17	RD14	15/14	33°49.720′	119°15.779′	5.4	43.2	64.9	5.0	30.8	125.0	3.9	6.2
Baked-contact test: granitic host rocks to dyke RD08												
18	<0.55 m	3/3	33°36.485′	119°53.237′	158.7	61.9	81.6	13.7	70.9	353.7	16.5	21.2
19	>0.55 m	11/7	33°36.485′	119°53.237′	315.9	−65.0	14.6	16.3	54.4	354.4	21.2	26.3

N/n = number of demagnetized/used samples; Slat, Slong = sample locality latitude and longitude; Decl, Incl = site mean declination, inclination; k = precision parameter of Fisher (1953); α_{95} = semi-angle of the 95% cone of confidence; Plat, Plong = latitude, longitude of the paleopole; Dp, Dm = semi-axes of the cone of confidence about the pole at the 95% probability level.

[a] Baked host rocks included.
[b] Chilled zone only.
[c] Excluded from the overall mean.
[d] Central part of the dyke.

Entry 18 – granite samples collected close than 0.55 m from the southern margin of RD08.
Entry 19 – granite samples collected farther than 0.55 m from the southern margin of RD08.

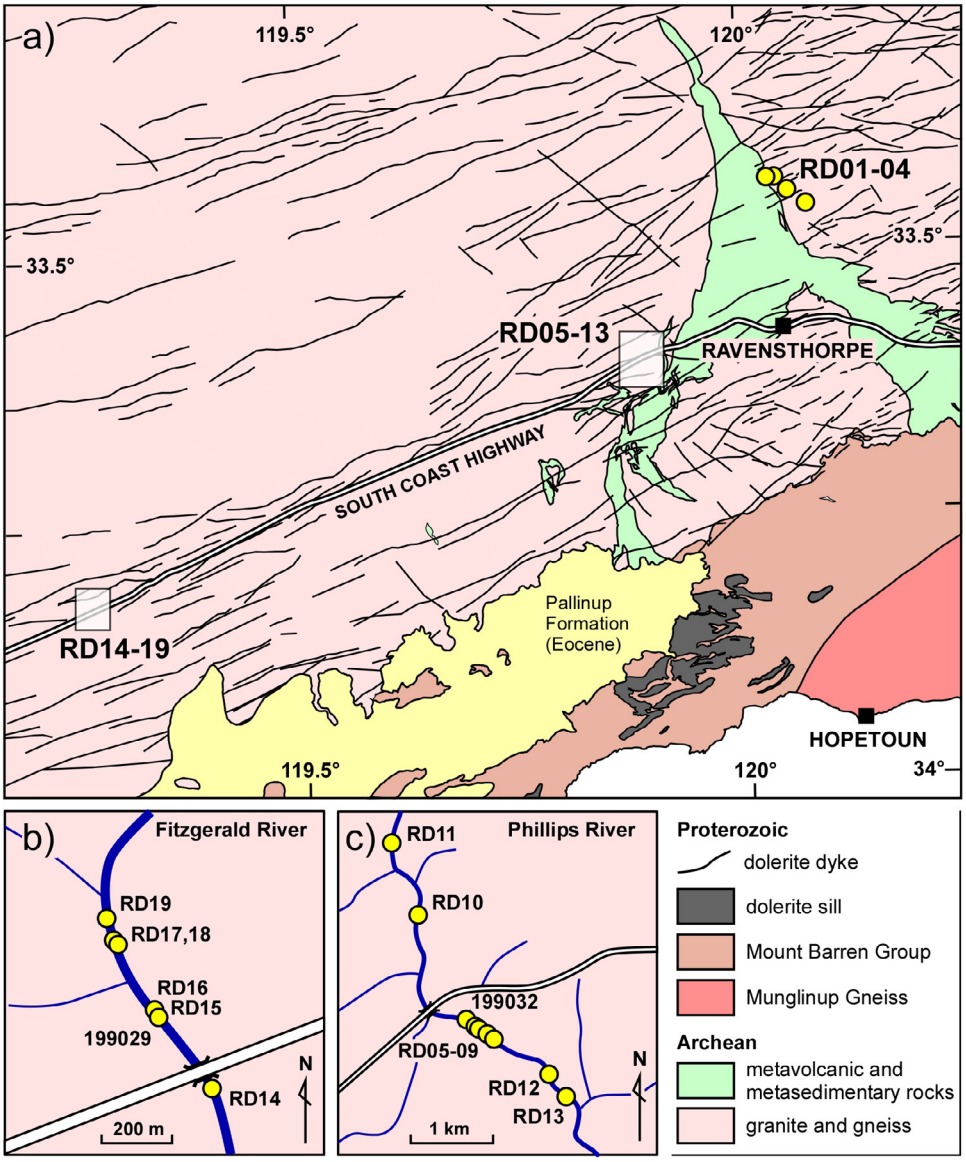


Fig. 2. (a) Geology of the region near Ravensthorpe, Western Australia, showing three areas sampled in east-northeast-trending dykes of the Gnowangerup–Fraser Dyke Suite. Southeast- and east-southeast-trending dykes belong to the Beenong and Nindibillup Dyke Suites, respectively (Spaggiari et al., 2009). (b) Locations of sampling sites are shown for dykes in the Fitzgerald River (a) and Phillips River (b). The Mount Barren Group and the Munglinup Gneiss are metamorphic components of the Albany–Fraser Orogen (Spaggiari et al., 2009, 2011).

(Baxter and Lipple, 1985) of the western Yilgarn Craton may be a good candidate). SHRIMP U–Pb ages of mafic dykes range from 1218 to 1202 Ma (Evans, 1999; Qiu et al., 1999; Wingate et al., 2000; Pidgeon and Nemchin, 2001; Pidgeon and Cook, 2003; Rasmussen and Fletcher, 2004; Wingate and Kirkland, 2011; MTD Wingate, unpublished data), with an average age of 1210 Ma. Emplacement of the Marnda Moorn LIP coincided with early Stage 2 (1215–1140 Ma) compressional tectonism and magmatism in the Albany–Fraser Orogen (Clark et al., 2000; Bodorkos and Clark, 2004; Spaggiari et al., 2009, 2011). The large areal extent (at least 0.5×10^6 km^2), short duration, and geochemical compositions of Marnda Moorn dykes may be consistent with a mantle plume origin (Wang et al., 2014), although the apparent concentration of dykes around the craton margins implies that plate boundary stresses were a major influence on dyke emplacement (Wingate and Pidgeon, 2005).

The Gnowangerup–Fraser Dyke Suite (Spaggiari et al., 2009, 2011) extends along the southern and southeastern margins of the Yilgarn Craton and the adjacent Albany–Fraser Orogen, and is part of the Marnda Moorn LIP. Within the Yilgarn Craton, dolerite dykes strike mainly east-northeast in the west, and northeast in the east, subparallel to the Albany–Fraser Orogen (Spaggiari et al., 2009). Myers, 1990 and Spaggiari et al., 2011 reported that some dykes become progressively recrystallized towards the craton margin and others are strongly deformed within the orogen, implying that at least some dykes were emplaced prior to the last major deformation (Stage 2) in the Albany–Fraser Orogen. Spaggiari et al., 2011 also noted that northeast-trending dykes in the northeastern part of the Albany–Fraser Orogen appear in aeromagnetic images to crosscut structures within the orogen; these dykes could constitute a different suite, or could still belong to the 1210 Ma Gnowangerup–Fraser Dyke Suite if this part of the orogen was not affected by Stage 2 events.

Two Gnowangerup dykes yielded zircon dates of 1203 ± 15 Ma and ~1238 Ma (Evans, 1999), although the latter result is based on only two analyses, and a dyke in the Stirling Ranges yielded a

zirconolite date of 1218 ± 3 Ma, in agreement with less precise dates for zircon and baddeleyite in the same sample (Rasmussen and Fletcher, 2004). Only preliminary paleomagnetic measurements are available for rocks of the Marnda Moorn LIP, owing to poor exposure and intense weathering. A dyke of the Fraser swarm, exposed in mine workings south of Kambalda, yielded a baddeleyite date of 1212 ± 10 Ma, and a steep downwards-directed magnetization shown to be primary by a detailed positive baked-contact test (Wingate et al., 2000; Pisarevsky et al., 2003b). The well-dated VGP (virtual geomagnetic pole) for this dyke, together with paleomagnetic data from c. 1205 Ma metamorphic rocks of the Albany–Fraser orogen, reliably places Australia at high latitudes at c. 1210 Ma (Wingate et al., 2002a; Pisarevsky et al., 2003b). Preliminary paleomagnetic results for four Gnowangerup dykes in the bed of the Phillips River also indicated steep, downwards-directed magnetizations (Giddings, 1976).

We collected paleomagnetic samples from 19 dykes (Table 1; Fig. 2), exposed along the Phillips and Fitzgerald Rivers, and north of Ravensthorpe. Samples for paleomagnetic study were collected using a portable diamond drill and oriented using solar and magnetic compasses. Coordinates of each site were determined using a GPS receiver. Samples for geochronology were collected from two of the sites (Table 1) also sampled for paleomagnetism. Geochemical results for the same rocks sampled for paleomagnetism are reported by Wang et al., 2014.

3. Analytical methods

3.1. Paleomagnetism

Each 23 mm diameter drillcore sample was cut into two or more 25 mm long specimens. Remanence composition was determined by both alternating-field (AF) and thermal demagnetization. Detailed stepwise AF demagnetization (up to 26 steps, ≤160 mT) employed a 2G-600 automated degaussing system and a 2G-755R cryogenic magnetometer in the paleomagnetic laboratory at the University of Western Australia. Thermal demagnetizations (up to 10 steps, ≤600 °C) were conducted using an ASC TD-48 thermal demagnetizer and a ScT cryogenic magnetometer in the Black Mountain paleomagnetic laboratory at The Australian National University. Magnetization vectors were isolated using Principal Component Analysis (Kirschvink, 1980). Curie temperature spectra and hysteresis properties were studied using a Petersen Instruments Variable Field Translation Balance (VFTB).

3.2. Geochronology

Each sample was crushed using a jaw crusher, powdered using a ring mill, and sieved. Zircon and baddeleyite were concentrated using conventional density and magnetic methods, and isolated by hand-picking. Zircons were mounted in 25 mm diameter epoxy-resin disks with chips of zircon standard BR266 (559 Ma, 903 ppm U; Stern, 2001) and crystals of zircon $^{207}Pb/^{206}Pb$ reference OGC1 (3465 Ma; OG1 of Stern et al., 2009). Baddeleyite crystals were mounted with chips of baddeleyite U/Pb standard PBR2 (2060 Ma Phalaborwa Carbonatite; Heaman, 2008) and crystals of zircon reference CZ3 as a uranium concentration standard (551 ppm ^{238}U; Pidgeon et al., 1994). Samples were documented with transmitted- and reflected-light optical microphotography, and zircons were also imaged with a cathodoluminescence (CL) detector fitted to a Philips XL30 scanning electron microscope (SEM) at the Department of Imaging and Applied Physics at Curtin University.

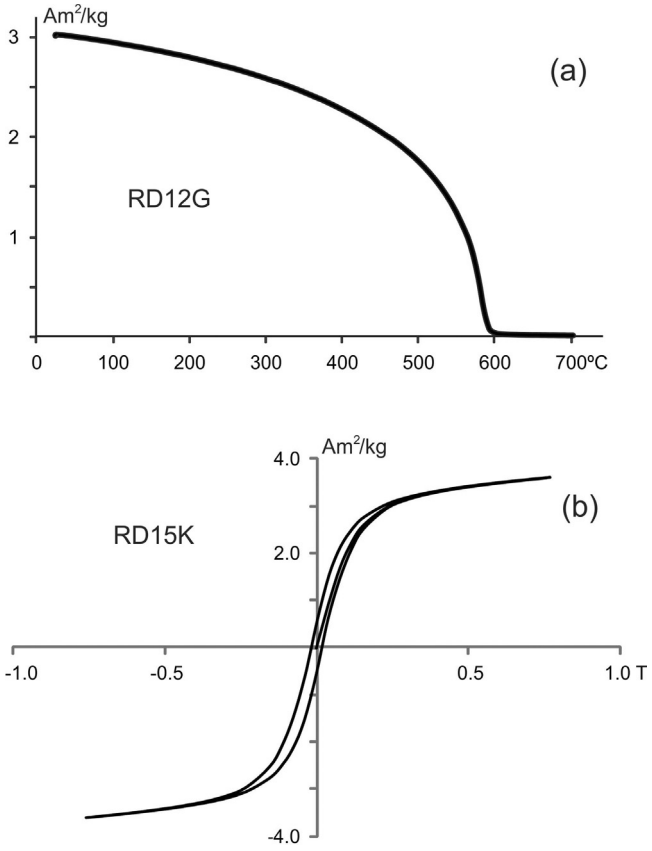


Fig. 3. Typical saturation magnetization versus temperature curve (RD12) (a) and typical hysteresis loop (RD15) (b).

U–Pb analyses were performed using the sensitive high-resolution ion microprobe (SHRIMP II) at Curtin University, using standard operating procedures similar to those described by Wingate and Kirkland, 2013. The diameter of the primary ion probe was about 20 μm for zircon and 15 μm for baddeleyite. Six data collection cycles (scans) were performed per analysis, and count times (per scan) were 10 s for the ^{204}Pb, ^{206}Pb, and ^{208}Pb mass peaks and background, and 30 s for the ^{207}Pb mass peak. Analyses of unknown zircons were referenced to multiple analyses of the BR266 standard for U/Pb calibration. Calibration and reproducibility uncertainties are included in the uncertainties of $^{238}U/^{206}Pb^*$ ratios (Pb^* = radiogenic Pb). Owing to crystal orientation effects that can bias $^{238}U/^{206}Pb^*$ ratios (Wingate and Compston, 2000), dates for baddeleyite are based on $^{207}Pb^*/^{206}Pb^*$ ratios. Data reduction was carried out using the software SQUID 2.50 and Isoplot 3.71 (MS Excel add-ins by Ludwig, 2003, 2009). Corrections for common Pb assumed an average crustal composition (Stacey and Kramers, 1975) appropriate to the age of the mineral. Mean ages are quoted below with 95% uncertainties ($t\sigma\sqrt{}$MSWD; Ludwig, 2003).

4. Petrography and rock magnetism

Most east-northeast-trending dykes (Group 1 below) are petrographically similar. The majority of dykes consist of medium- to coarse-grained quartz dolerite or gabbro and exhibit intergranular to subophitic textures. These samples contain 40–50% euhedral plagioclase, 20–30% interstitial quartz and quartz-feldspar granophyre, 20–30% orthopyroxene (enstatite) and clinopyroxene (augite ± pigeonite) partially altered to chlorite and blue-green

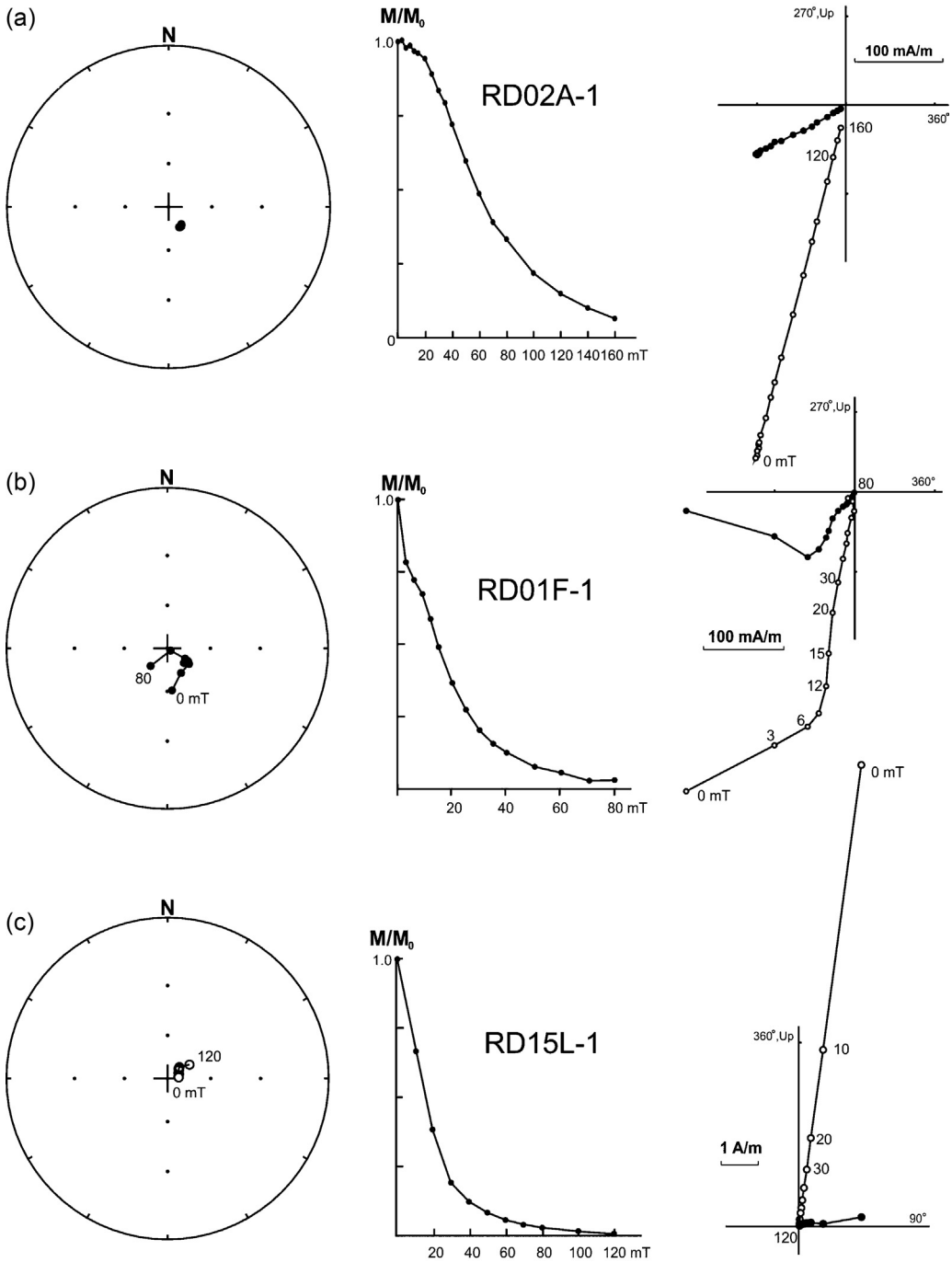


Fig. 4. Examples of alternating-field (AF) demagnetization of Group 1 dykes. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots (equal-angle, lower-hemisphere projection) show upwards (downwards) pointing paleomagnetic directions with open (closed) symbols.

amphibole, and minor iron-titanium oxide minerals, epidote, biotite, and apatite. Plagioclase in many samples is strongly altered to saussurite. Some samples contain radial plagioclase glomerocrysts, coarse-grained poikilitic clinopyroxene, and minor olivine. Iron oxide minerals are typically euhedral and cubic.

A typical thermomagnetic curve and a hysteresis loop are shown in Fig. 3 for samples of dykes at sites RD12 and RD15. The Curie temperatures are close to that of magnetite (580 °C). The M_{rs}/M_s ratio (where M_s is the saturation magnetization and M_{rs} is the saturation remanence) varies from 0.21 to 0.36, and the B_{cr}/B_c ratio (where B_c is the coercivity and B_{rc} is the coercivity of remanence) varies from 1.5 to 2.6 for most of the studied dykes. These values are characteristic of paleomagnetically stable, pseudo-single-domain (PSD) magnetite (Day et al., 1977). One exception is the coarse-grained central part of the RD12 dyke, for which values of $M_{rs}/M_s = 0.07$ and $B_{cr}/B_c = 5$ suggest the dominance of low-coercivity, multi-domain (MD) magnetite as the remanence carrier. This supports the suggestion, made below, that the remanence of the central part of the RD12 dyke is a secondary remagnetization.

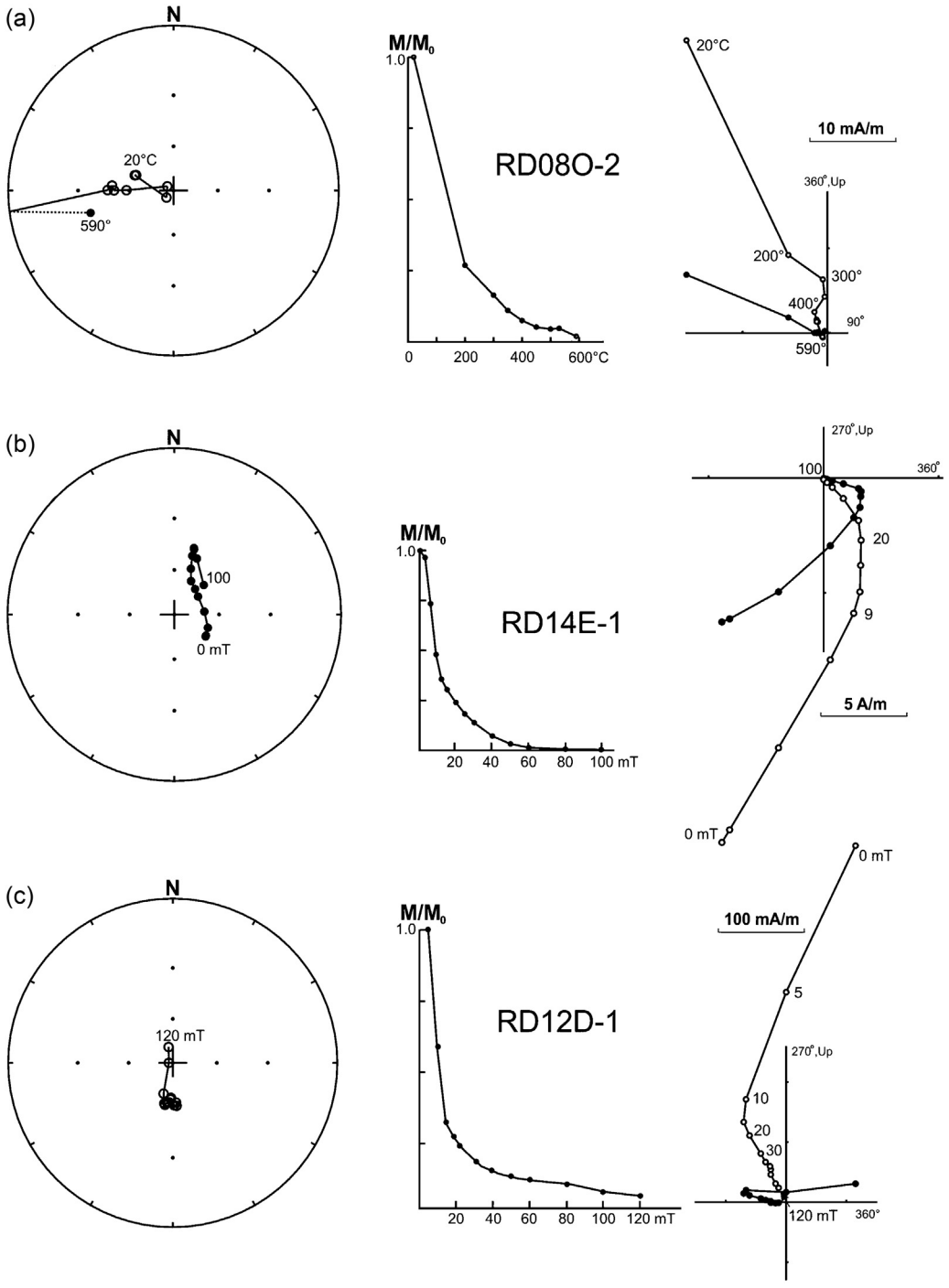


Fig. 5. Examples of thermal and AF demagnetization of the unbaked host rocks near the RD08 dyke (a), dyke RD14 (Group 2) (b) and the central part of dyke RD12 (Group 2) (c). See Fig. 4 caption for other notes.

5. Paleomagnetism

Based on paleomagnetic characteristics, the paleomagnetic results for the studied dykes are divided into three groups (Table 1).

5.1. Group 1

Samples in this group represent east-northeast-trending dykes of the Gnowangerup swarm. Alternating-field (AF) demagnetization of 15 dykes reveals a stable characteristic magnetization (ChRM), directed steeply downwards (Table 1). In most cases, this is the only component of natural remanent magnetization (NRM) (Fig. 4a), although there are indications of a low-stability, randomly oriented, probably viscous component (Fig. 4b). The ChRM direction in 13 of 15 dykes is steeply downward towards the southeast (Table 1; Fig. 6a). One exception is dyke RD13, which shows a northeast declination (Table 1, entry 11), possibly reflecting significant geomagnetic secular variation at the time of its emplacement. This direction is regarded as anomalous and excluded from further interpretation.

The ChRM direction of dyke sample RD15 (Fig. 4c; Table 1, entry 12) is antipodal to those of the others in Group 1, suggesting that the duration of dyke emplacement was sufficiently long to have accommodated at least one geomagnetic reversal (Table 1;

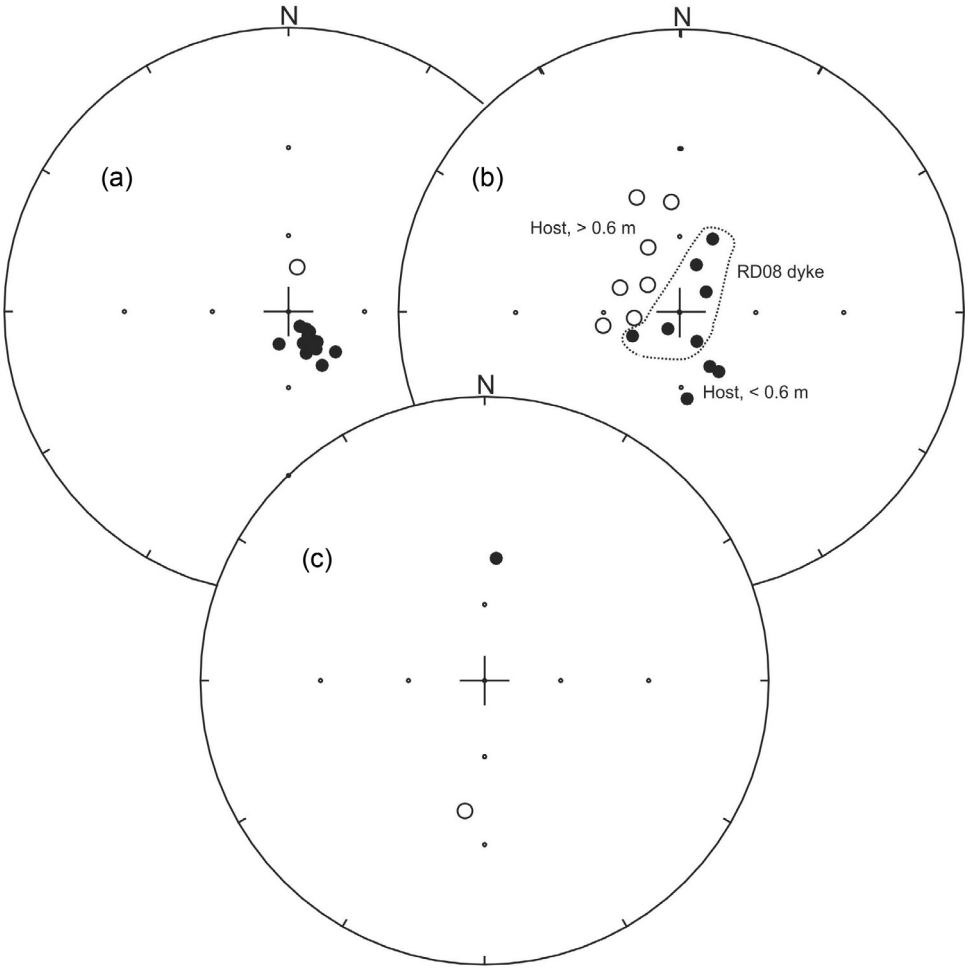


Fig. 6. Stereoplots (equal-angle projection) of: (a) site mean directions of Group 1dykes (Table 1, entries 1–10, 12–15); (b) sample directions of the RD08 dyke and host rocks near its contact; (c) site mean directions of Group 2 dykes.

Figs. 4c and 6a). The reversal test of McFadden and McElhinny, 1990 is positive at the sample level with classification R_c ($\Delta = 12.9^\circ$, $\Delta_c = 15.7^\circ$).

The east-northeast-trending, 10 m thick RD12 dyke exhibits unusual paleomagnetic characteristics. Four samples from its southern chilled margin yield a ChRM direction (Table 1, entry 10), which is similar to those of the other Group 1 samples, and is included with Group 1. However, eight samples from the centre of the dyke yield a ChRM direction (Table 1, entry 16) almost exactly antipodal to that of the southeast-trending RD14 (Group 2) dyke (Table 1, entry 17; Fig. 6c). These results are discussed further below.

To test whether the Group 1 characteristic remanence (ChRM) is primary, we collected host-rock samples adjacent to the 2 m thick RD08 dyke for a baked-contact test. In addition to the dyke samples, the host granitic gneiss was sampled at distances between 18 cm and 15.3 m from the southern dyke contact. Thermal and AF demagnetization of three samples reveals a steep southeast-downward direction (Table 1, entry 18) in gneiss within 55 cm of the dyke contact. This implies that the host rocks adjacent to the dyke were reheated and remagnetized at the time of dyke emplacement. Seven of 11 gneiss samples collected more than 55 cm from the dyke contact yield a consistent steep, northwest-upwards direction (Table 1 entry 19; Figs. 5a and 6b). We interpret this as a positive baked-contact test that suggests the ChRM of the dyke is primary. Although the mean directions of the unbaked host rocks and the RD08 dyke are nearly antipodal (Table 1), the reversal test of McFadden and McElhinny, 1990 is negative. The remanence of the unbaked host rocks was probably acquired during the earlier stages of the Albany–Fraser Orogeny, prior to the emplacement of the dykes.

A positive baked-contact test, the presence of antipodal magnetization directions, and rock magnetic evidence for the dominance of stable PSD magnetite, all lead to the conclusion that the characteristic Group 1 magnetization ($D = 149.0^\circ$, $I = 74.8^\circ$, $k = 122.3$, $\alpha_{95} = 6.5^\circ$) is primary. In addition, this characteristic direction is almost identical to the ChRM direction of the 1212 Ma Fraser swarm dyke ($D = 152.6^\circ$, $I = 74.1^\circ$, $k = 110$, $\alpha_{95} = 2.9^\circ$), which was also shown to be primary by a highly detailed and robust baked-contact contact test (Wingate et al., 2000; Pisarevsky et al., 2003b).

5.2. Group 2

The southeast-trending, very fine grained RD14 dyke, exposed in the Fitzgerald River, yields a paleomagnetic direction (Table 1, entry 17; Fig. 5b) similar to that of the 755 Ma Mundine Well Dyke Suite of the Gascoyne Province and Pilbara Craton ($D = 14.2^\circ$, $I = 36.4^\circ$, $\alpha_{95} = 7.5^\circ$, Wingate and Giddings, 2000). This dyke possibly belongs to the east-southeast-trending Nindibillup Dyke Suite or the southeast-trending Beenong Dyke Suite, both of which crosscut major structures of the Albany–Fraser Orogen and clearly postdate Stage 2 of the orogeny (Spaggiari et al., 2009). SHRIMP analysis of a single baddeleyite crystal from a Nindibillup dyke east of the

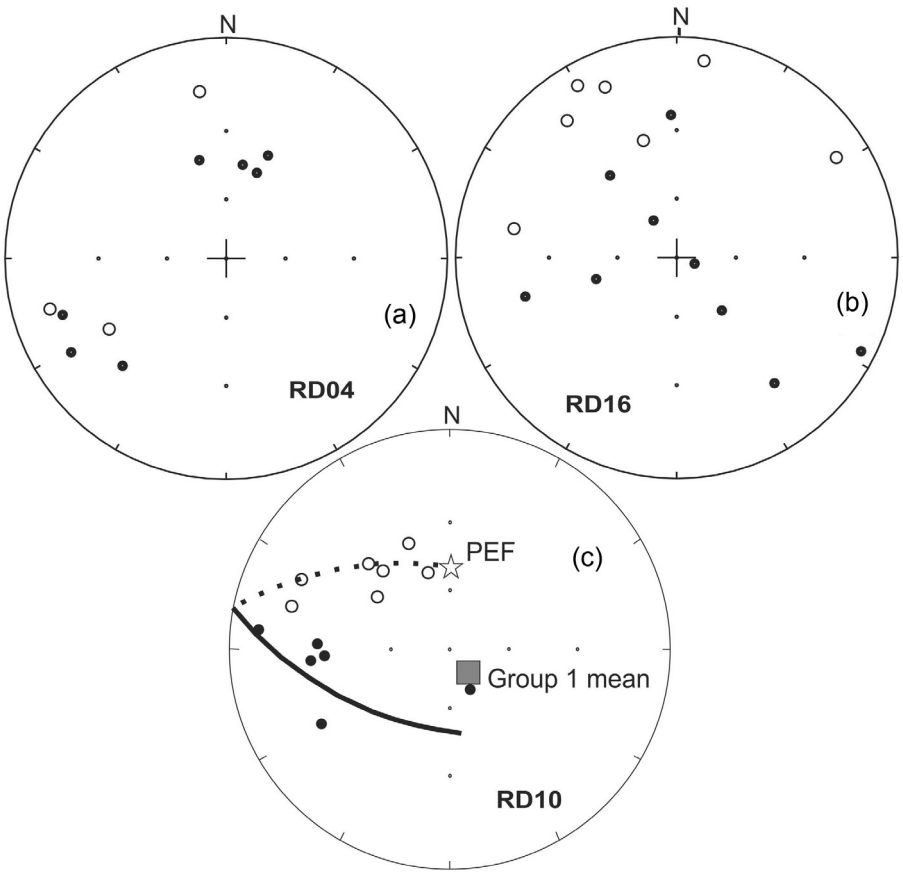


Fig. 7. Stereoplots (equal-angle projection) of sample directions of Group 3 dykes. In (c), the overall mean direction for Group 1 dykes in shown; PEF = direction of the present Earth's field in the study area.

Lort River indicates a preliminary date of c. 750 Ma (MTD Wingate, unpublished data). Together with the paleomagnetic direction, this suggests that the Nindibillup dykes are coeval with the 755 Ma Mundine Well dykes. Geochemical characteristics of the RD14 dyke are distinct from those of the Group 1 dykes (Wang et al., 2014) and the trace element distribution patterns are similar to those of the Mundine Well dykes (Li et al., 2006).

Samples from the centre of the RD12 dyke yield a ChRM direction (Table 1, entry 16; Fig. 5c and 6c) that is antipodal to the Group 2 ChRM direction (Table 1, entry 17). However, as mentioned above, the chilled margin of the RD12 dyke carries a typical Group 1 ChRM direction (Table 1, entry 10). The most likely explanation is that the coarser-grained interior of the dyke, which is dominated by low-stability MD magnetite (see Section 4) was remagnetized during the younger (Nindibillup or Beenong) event, whereas the more stable, finer-grained magnetite in the chilled margin was unaffected.

5.3. Group 3

Three east-northeast-trending dykes RD04 (33°27.394′ S, 120°06.387′ E), RD10 (33°35.609′ S, 119°52.616′ E) and RD16 (33°49.606′ S, 119°15.673′ E) yield anomalous remanence directions (Fig. 7). Unlike the dykes in Groups 1 and 2, the RD04 dyke is not exposed in a river bed, and is strongly weathered. Remanence directions form two dispersed components (Fig. 7a), one of which (north-downward) is close to the mean remanence direction of the RD14 dyke, and might be associated with the younger (c. 755 Ma?) magmatic event. The origin of the second component, directed shallowly to the southwest, is unknown. Remanence directions in the RD16 dyke are randomly oriented (Fig. 7b). Remanence directions in the RD10 dyke (Fig. 7c) are distributed along a great circle between the mean Group 1 direction and the direction of the present-day dipole geomagnetic field (present Earth's field, PEF). The results for these three dykes are excluded from further discussion.

6. Geochronology

Samples for geochronology were collected from the same outcrops sampled for paleomagnetism in two east-northeast-trending Group 1 dykes. Sample 199029 corresponds to dyke RD15 in the Fitzgerald River, and sample 199032 to dyke RD05 in the Phillips River.

Sample 199029 is a medium- to coarse-grained gabbro, which yielded colourless, irregular to subhedral zircons up to 200 μm long. In cathodoluminescence (CL) images (Fig. 8b), the crystals are either internally featureless, or exhibit broad oscillatory zoning. Eight analyses of the BR266 zircon standard indicated an external spot-to-spot (reproducibility) uncertainty of 0.00% (a minimum value of 0.50% (1σ) was assumed and propagated to the unknown analyses) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.15% (1σ). Twenty-four analyses were obtained from 18 zircons (Table 2, Fig. 8a). The 204-corrected $^{207}Pb^*/^{206}Pb^*$ ratios correlate with their common-Pb corrections, indicating that corrections using ^{204}Pb are inaccurate for some or all of these analyses. The age of the sample is therefore determined using a common-Pb regression, although common Pb is sufficiently low that neither the form of common-Pb correction, nor its application, has a significant effect on calculated dates. A regression from initial $^{207}Pb/^{206}Pb$ (0.9257 at 1210 Ma) through 18 analyses, uncorrected for common Pb, yields a

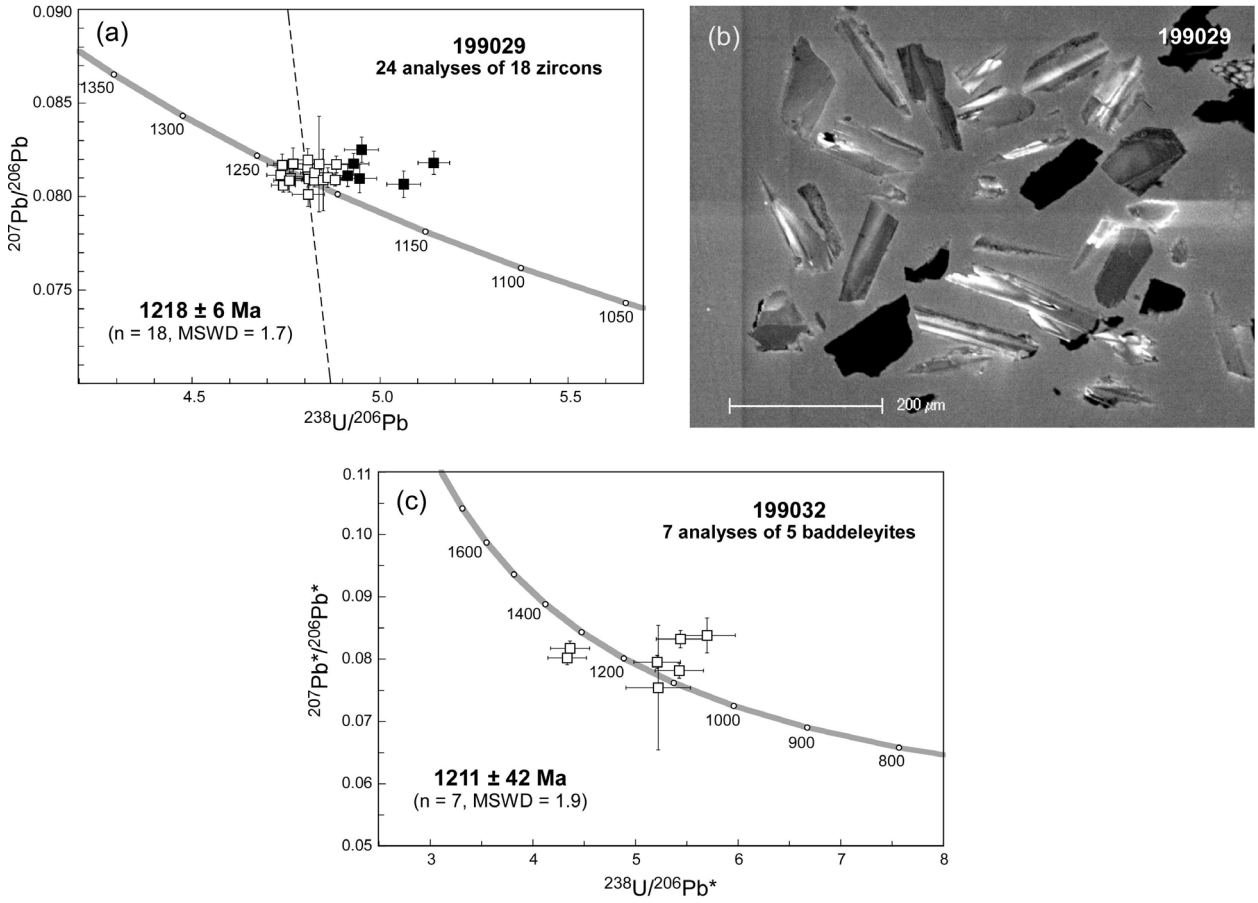


Fig. 8. (a) U–Pb analytical data for zircons from sample 199029. (b) Cathodoluminescence (CL) image of zircons in sample 199029. (c) U–Pb analytical data for baddeleyites from sample 199032. In (a), the data are not corrected for common Pb, black squares indicate analyses interpreted to reflect radiogenic-Pb loss, and the dashed line indicates a regression from initial Pb.

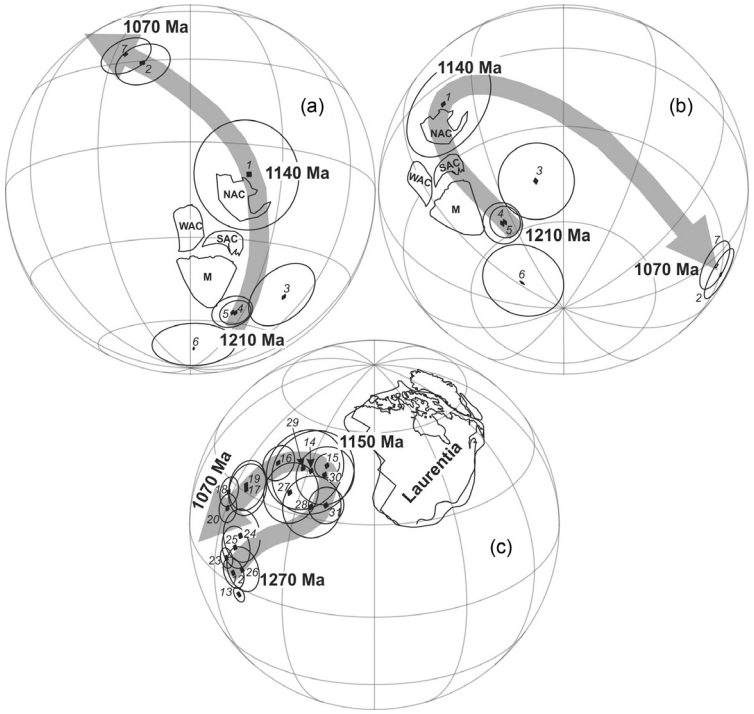


Fig. 9. Late Mesoproterozoic Apparent Polar Wander Paths (APWPs) for: (a and b) Australia, with two polarity options (see text); (c) Laurentia. WAC, NAC, SAC, M = West, North, South Australian, Mawson Cratons. Pole numbers refer to entries in Table 3.

Table 2
U-Pb analytical data for zircon and baddeleyite from Gnowangerup–Fraser dyke samples 199029 and 199032.

Grain Area	^{238}U (ppm)	^{232}Th (ppm)	Th/U	f (%)	^{238}U/^{206}Pb	(±1σ)	^{207}Pb/^{206}Pb	(±1σ)	^{238}U/^{206}Pb*	(±1σ)	^{207}Pb/^{206}Pb*	(±1σ)	^{238}U/^{206}Pb* age (Ma)	(±1σ)	^{207}Pb*/^{206}Pb* age (Ma)	(±1σ)	Disc. (%)
Sample 199029 (zircon), dyke RD15																	
1.1	234	784	3.46	0.143	4.759	0.041	0.08086	0.00061	4.766	0.041	0.07965	0.00075	1228	10	1188	18	−3.3
2.1	910	1411	1.60	0.016	4.849	0.033	0.08089	0.00164	4.850	0.033	0.08075	0.00164	1209	7	1215	40	0.6
3.1	332	354	1.10	0.076	4.807	0.037	0.08146	0.00053	4.810	0.037	0.08082	0.00059	1218	9	1217	14	0.0
*4.1	190	104	0.57	−0.095	5.063	0.045	0.08066	0.00071	5.058	0.045	0.08146	0.00081	1163	9	1233	20	5.6
5.1	293	160	0.56	−0.026	4.861	0.038	0.08102	0.00052	4.860	0.038	0.08124	0.00054	1206	9	1227	13	1.7
*6.1	271	314	1.20	0.157	5.143	0.042	0.08180	0.00062	5.151	0.042	0.08047	0.00077	1144	9	1209	19	5.4
*7.1	190	152	0.83	0.066	4.951	0.045	0.08250	0.00068	4.955	0.045	0.08194	0.00075	1185	10	1244	18	4.7
8.1	309	280	0.93	0.044	4.809	0.039	0.08106	0.00055	4.811	0.039	0.08069	0.00059	1217	9	1214	14	−0.3
9.1	479	452	0.98	0.047	4.884	0.035	0.08172	0.00045	4.886	0.035	0.08132	0.00048	1200	8	1229	12	2.3
*10.1	273	372	1.41	0.307	4.930	0.040	0.08175	0.00057	4.945	0.040	0.07915	0.00082	1187	9	1176	20	−1.0
11.1	388	345	0.92	0.047	4.826	0.037	0.08127	0.00050	4.828	0.037	0.08087	0.00054	1213	8	1218	13	0.4
12.1	815	1376	1.74	−0.040	4.762	0.032	0.08097	0.00035	4.760	0.032	0.08131	0.00037	1229	8	1229	9	0.0
13.1	862	1591	1.91	0.000	4.743	0.032	0.08060	0.00035	4.743	0.032	0.08060	0.00035	1233	8	1212	9	−1.8
14.1	1071	1733	1.67	0.009	4.881	0.032	0.08089	0.00031	4.882	0.032	0.08082	0.00031	1201	7	1217	8	1.3
*15.1	213	128	0.62	0.027	4.946	0.045	0.08097	0.00077	4.948	0.045	0.08075	0.00081	1187	10	1215	20	2.4
16.1	373	274	0.76	−0.067	4.736	0.037	0.08115	0.00054	4.733	0.037	0.08172	0.00060	1236	9	1239	14	0.3
17.1	119	73	0.63	0.067	4.769	0.051	0.08175	0.00085	4.772	0.051	0.08118	0.00094	1226	12	1226	23	−0.1
*18.1	278	198	0.74	0.374	4.914	0.040	0.08113	0.00059	4.933	0.041	0.07797	0.00089	1190	9	1146	23	−3.8
14.2	235	139	0.61	−0.018	4.838	0.041	0.08174	0.00256	4.837	0.041	0.08190	0.00257	1211	9	1243	61	2.5
16.2	243	167	0.71	0.032	4.740	0.040	0.08168	0.00059	4.742	0.040	0.08140	0.00062	1234	10	1231	15	−0.2
1.2	210	621	3.06	0.065	4.809	0.043	0.08012	0.00067	4.812	0.043	0.07958	0.00074	1217	10	1187	18	−2.6
2.2	846	1255	1.53	0.069	4.811	0.087	0.08094	0.00035	4.814	0.087	0.08036	0.00039	1217	20	1206	9	−0.9
13.2	853	1531	1.85	0.024	4.825	0.033	0.08087	0.00036	4.826	0.033	0.08067	0.00037	1214	8	1213	9	0.0
15.2	381	297	0.80	0.083	4.808	0.039	0.08196	0.00060	4.812	0.039	0.08125	0.00068	1217	9	1228	16	0.8
Sample 199032 (baddeleyite), dyke RD05																	
1.1	138	11	0.08	0.458	5.425	0.249	0.08442	0.00130	5.440	0.238	0.08321	0.00139	1088	44	1274	33	11.2
2.1	131	4	0.03	0.889	4.354	0.200	0.08235	0.00116	4.363	0.190	0.08172	0.00120	1330	52	1239	29	−26.9
3.1	27	5	0.20	2.931	5.656	0.277	0.08911	0.00250	5.699	0.274	0.08381	0.00281	1042	46	1288	65	−43.1
4.1	170	41	0.25	0.332	5.180	0.295	0.08102	0.00087	5.222	0.315	0.07541	0.01000	1129	62	1079	266	1.9
5.1	211	21	0.10	0.217	5.185	0.236	0.08281	0.00096	5.213	0.227	0.07954	0.00106	1131	45	1185	26	7.8
2.2	138	8	0.06	0.404	4.326	0.198	0.08077	0.00102	4.335	0.188	0.08020	0.00111	1338	52	1202	27	−20.0
1.2	165	11	0.07	0.387	5.406	0.247	0.08021	0.00115	5.427	0.236	0.07816	0.00122	1090	44	1151	31	3.0

Notes: Analyses marked with asterisks were not included in calculation of the mean age for the sample. f is the proportion of common ^{206}Pb in total measured ^{206}Pb. Pb* = radiogenic Pb. Radiogenic ratios and ages are corrected for common Pb. Disc. (discordance, in %) = 100 × ((^{207}Pb/^{206}Pb date) − (^{238}U/^{206}Pb date))/(^{207}Pb/^{206}Pb date). U/Pb ratios and ages include calibration uncertainties.

concordia intercept of 1218 ± 6 Ma (MSWD = 1.7), interpreted as the age of magmatic crystallization. Six analyses yield significantly younger 207-corrected $^{238}U/^{206}Pb^*$ dates of 1192–1140 Ma, interpreted to reflect minor loss of radiogenic Pb.

Sample 199032 is also a medium- to coarse-grained gabbro. The 3 kg sample yielded seven pleochroic brown, subhedral, highly fractured baddeleyite crystals up to 100 μm long. Twelve analyses of the PBR2 baddeleyite standard indicated an external spot-to-spot (reproducibility) uncertainty of 4.30% and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 1.27% (1σ). Seven analyses were obtained from five crystals (Table 2; Fig. 8c), and yield a 208-corrected (Compston et al., 1984; Wingate et al., 1998) weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 1211 ± 42 Ma (MSWD = 1.9). Although one of the analyses (4.1) is imprecise, excluding it has no effect on the calculation. The date of 1211 ± 42 Ma is interpreted as the age of magmatic crystallization of the dyke.

7. Paleogeographic implications

There is general agreement that Australia was assembled by the amalgamation of the North, West, and South Australian Cratons (NAC, WAC, and SAC), though the timing of this assembly is still debated (e.g., Myers et al., 1996; Betts and Giles, 2006; Schmidt et al., 2006; Cawood and Korsch, 2008; Li and Evans, 2011; Schmidt, 2014). In this study we adopt the model of Li and Evans, 2011, which suggests proximity between the three cratons since c. 1800 Ma, but in a configuration different from that of the present-day until c. 650 Ma. According to this model, there was a 40° intraplate rotation between the NAC and WAC–SAC at 650–550 Ma which resulted in the present-day configuration.

The mean Group 1 paleomagnetic pole is almost identical to the primary VGP of the 1212 Ma Fraser dyke at the southeastern edge of the Yilgarn Craton (Table 3, entry 4; Fig. 1; Wingate et al., 2000; Pisarevsky et al., 2003b). The new paleopole for the Gnowangerup–Fraser dykes places Australia in a near-polar position at c. 1210 Ma.

The similarity of coeval fragments of APWPs of two continents is a necessary condition for, but not conclusive proof of, coherent drift of the two continents within a stable supercontinent. On the other hand, significant differences between two coeval APWPs in either their lengths or shapes would preclude their belonging to the same supercontinent.

Segments of Australian and Laurentian APWPs, based on poles from Table 3, are shown in Fig. 9. Because the 1140 and 1070 Ma Australian poles are more than 50° apart, Fig. 9a and b show two versions of the Australian APWP, according to the two polarity options for the 1070 Ma poles. However, both Australian APWPs are significantly dissimilar to the corresponding segments of the Laurentian APWP in length and shape (Fig. 9c), suggesting that Australia and Laurentia moved independently between 1210 and 1070 Ma and were not parts of a single supercontinent. Our new pole supports the conclusions of Wingate et al., 2002a and Pisarevsky et al., 2003b, i.e., that none of the previously suggested Mesoproterozoic reconstructions juxtaposing Australia and Laurentia, e.g. SWEAT (Moores, 1991; Dalziel, 1991; Hoffman, 1991), AUSWUS (Brookfield, 1993; Karlstrom et al., 1999; Burrett and Berry, 2000), or AUSMEX (Wingate et al., 2002b), is paleomagnetically permissible at c. 1210 Ma. Note that the AUSMEX reconstruction was proposed only for c. 1070 Ma.

An analysis of 1800–1580 Ma geological and paleomagnetic data from North Australia suggested the possibility of a "SWEAT-like" reconstruction, with North Australia located close to the northwestern tip of Laurentia (Betts et al., 2008). This suggestion does not contradict Mesoproterozoic paleomagnetic data (e.g., Zhang et al., 2012). Combining geological and paleomagnetic

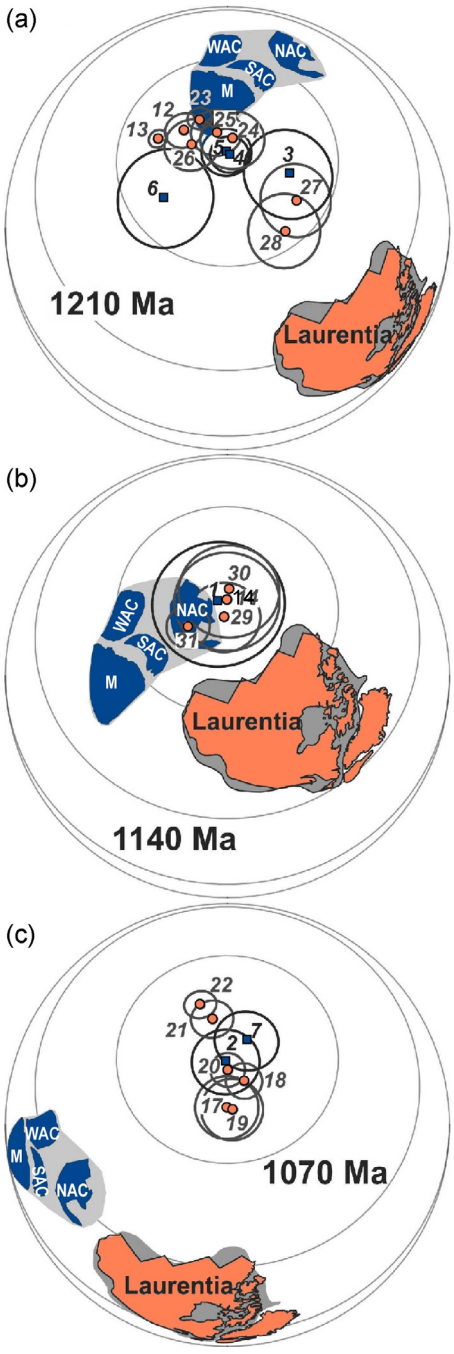


Fig. 10. Paleomagnetically permissible reconstructions of Australia and Laurentia at 1210–1070 Ma. WAC, NAC, SAC, M = West, North, South Australian, Mawson Cratons. Pole numbers refer to entries in Table 3. Euler rotation parameters to absolute framework; each consists of Euler pole latitude, longitude, rotation (+anticlockwise): (a) Laurentia: (50.86, −142.22, −143.55), NAC: (20.07, −150.76, −149.47); (b) Laurentia: (66.53, −132.90, −151.65), NAC: (21.15, −164.56, −106.98); (c) Laurentia: (53.54, −178.12, −174.22), NAC: (19.75, 146.89, −95.86). WAC and SAC rotated to NAC as in Li and Evans, 2011. Mawson rotated to SAC: (1.30, 37.71, 30.31).

evidence, Pisarevsky et al., 2014 proposed an internally consistent, animated paleogeographic model, in which Australia collided with western Laurentia at c. 1650–1600 Ma, during the Racklan Orogeny (Furlanetto et al., 2013) and broke away between 1450 and 1380 Ma. Following this model, we propose c. 1210–1070 Ma reconstructions of Australia and Laurentia (Fig. 10). Being widely separated at c. 1210 Ma (Fig. 10a), the two continents came closer at c. 1140 Ma (Fig. 10b). At c. 1070 Ma, the Alcurra and Bangemall Sills paleopoles (Table 3, entries 2 and 7) permit an AUSMEX-like

Table 3
Paleomagnetic poles for c. 1300–1050 Ma.

#	Pole name	Age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	Refs.
North Australia						
1	Mt. Isa dolerite dykes	1141–1139	9.5	311.1	17.4	Tanaka and Idnurm, 1994
2	Alcurra intrusions	1089–1064	2.8	80.4	8.8	Schmidt et al., 2006
West Australia						
3	Mt Barren Group	c. 1205	46.6	347.4	12.0	Pisarevsky et al., 2003b; Dawson et al., 2003
4	Fraser dyke VGP	1212 ± 13	55.8	325.7	5.0	Wingate et al., 2000; Pisarevsky et al., 2003b
5	Gnowangerup dykes	c. 1210	55.8	323.9	6.5	This study
6	Bremer Bay meta-morphic rocks (BB1)	c. 1200	74.4	303.8	13.0	Pisarevsky and Harris, 2001; Pisarevsky et al., 2003b
7	Bangemall sills (BBS)	1070 ± 6	33.9	95.0	8.3	Wingate et al., 2002b
Baltica						
8	Post-Jotnian intrusions	1270–1246	−1.8	159.1	3.4	Elming and Pesonen, 2010
9	Salla dyke VGP	1129–1115	71.0	113.0	8.0	Salminen et al., 2009
10	Bamble intrusions	1100–1040	−15.1	222.5	19.3	Elming and Pesonen, 2010
11	Laanila dolerite	1095–995	−2.1	212.2	13.8	Mertanen et al., 1996
Laurentia						
11	Nain anorthosite	1320–1290	11.7	206.7	2.2	Murthy, 1978
12	Mackenzie dykes	c. 1270	4.0	190.0	5.0	Buchan and Halls, 1990 and LeCheminant and Heaman, 1989
13	Sudbury dykes	c. 1235	−2.5	192.8	2.5	Palmer et al., 1977
14	Abitibi dykes	1141 ± 2	44.4	211.4	13.7	Ernst and Buchan, 1993
15	Nipigon sills and lavas	c. 1110	47.2	217.8	4.0	Elming and Pesonen, 2010
16	Osler volcanics	c. 1107	43.1	194.5	5.8	Halls, 1974 and Davis and Green, 1997
17	Chengwatana volc.	1095 ± 2	30.9	186.1	8.2	Kean et al., 1997 and Zartman et al., 1997
18	Portage Lake volc.	1095 ± 3	26.7	178.0	4.7	Hnat et al., 2006 and Davis and Paces, 1990
19	Cardenas basalts	1091 ± 5	32.0	185.0	8.0	Weil et al., 2003
20	Lake shore traps	1087 ± 2	22.2	180.8	4.5	Diehl and Haig, 1994 and Davis and Paces, 1990
21	Nonesuch shale	c. 1050	7.6	178.1	5.5	Henry et al., 1977 and Wingate et al., 2002b
22	Freda sandstone	c. 1050	2.2	179.0	4.2	Henry et al., 1977 and Wingate et al., 2002b
Greenland						
23	Kungnat ring dyke	1275 ± 2	3.4	198.7	3.2	Piper and Stearn, 1977
24	ntr.	1275 ± 1	13.2	202.6	8.3	Piper, 1992
25		1251–1236	8.7	201.7	6.6	Piper and Stearn, 1977
26		1249–1227	3.2	206.4	7.2	"
27	Hviddal dyke	1184 ± 5	33.2	215.3	9.6	Piper, 1977; Upton et al., 2003
28	Narssaq gabbro	1184 ± 5	31.6	225.4	9.7	"
29	Giant gabbro dykes	1163 ± 2	42.3	226.1	9.4	Piper, 1977; Buchan et al., 2001
30	South Qoroq intr.	1163 ± 2	41.8	215.9	13.1	Piper, 1992
31	NE-trending dyke swarm	1165–1155	33.4	230.8	5.7	Piper, 1992
India						
32	Harohalli alk. dykes	1192 ± 10	24.9	78.0	15.0	Pradhan et al., 2008
33	Mahoba dykes	1113 ± 7	38.7	229.5	12.4	Pradhan et al., 2012
34	Majhgawan Kim. VGP	1174 ± 14	36.8	212.5	12.2	Gregory et al., 2006
Amazonia						
35	loresta Fm.	c. 1200	24.6	164.6	5.5	Tohver et al., 2002
36	Fortuna Fm.	1150 ± 7	59.8	155.9	9.0	DAgrella-Filho et al., 2008DAgrella-Filho et al., 2008

A_{95} = semi-angle of the 95% circle of confidence about the mean paleopole.

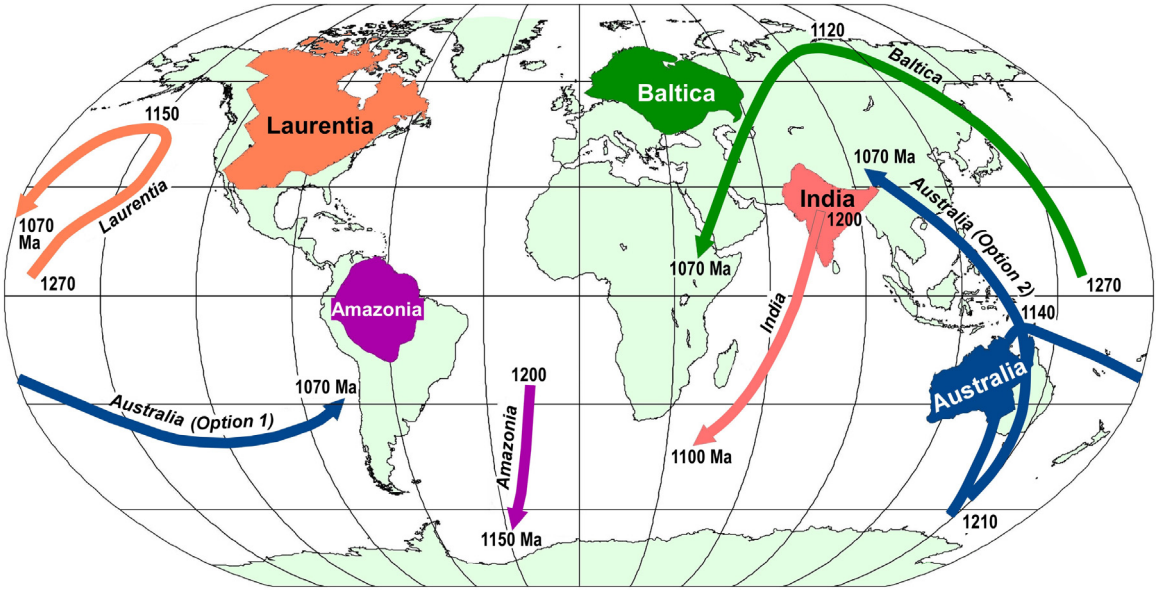


Fig. 11. Late Mesoproterozoic Apparent Polar Wander Paths (APWPs) for Australia, Laurentia, Baltica, Amazonia, and India.

reconstruction (Fig. 10c). Note that all three reconstructions are speculative owing to longitudinal uncertainty, but they minimize the relative movements of two continents.

The analysis of 1300–1070 Ma paleopoles from various continents (Table 3; Fig. 11) demonstrates that Baltica and Amazonia also drifted independently from Laurentia or Australia, because the angular distance between corresponding paleopoles varies from 17 to 98°. Elming et al., 2009 analyzed data from Laurentia and Amazonia, and demonstrated significant relative movement of these two continents between 1200 and 980 Ma. The c. 1270 Ma Laurentian poles are located quite close to c. 1070 Laurentian poles, whereas the angular distance between c. 1270 and 1070 Ma poles for Baltica is much greater (Fig. 11). Cawood et al., 2010 suggested a c. 95° clockwise rotation of Baltica with respect to Laurentia at the end of the Mesoproterozoic. However, the recent paleomagnetic result of Salminen et al., 2009 (Table 3, entry 9) suggests more complicated drift of Baltica with respect to Laurentia (Pisarevsky et al., 2013). The c. 1200–1150 Ma Australian and Indian paleopoles are consistent with a connection between them during this time, although this reconstruction should be carefully tested with geological data.

Fig. 11 clearly shows that between c. 1270 Ma and 1070 Ma most continents appear to have drifted independently. Importantly, paleomagnetic data for the main constituents of Rodinia, i.e., Laurentia, Baltica, Australia, and Amazonia, could not have been parts of a single supercontinent at that time. This supports the idea that the final assembly of Rodinia did not occur until the early Neoproterozoic (Li et al., 2008). We speculate that a large supercontinent did not exist in the late Mesoproterozoic. This may have been a transitional time between the final breakup of Nuna and the assembly of Rodinia.

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